speculate. My hunch is that one day we shall discover a dispersal phase which has hitherto, for some reason, been overlooked. It is too much to hope that the dispersing individuals will literally sprout wings! But they might in various ways be equipped for life above ground rather than underground. They could be hairy instead of naked, for instance. Naked mole rats don't regulate their individual body temperatures in the way that normal mammals do; they are more like 'cold-blooded' reptiles. Perhaps they control temperature socially—another resemblance to termites and bees. Or could they be exploiting the well-known constant temperature of any good cellar? At all events, my hypothetical dispersing individuals might well, unlike the underground workers, be conventionally 'warm-blooded'. Is it conceivable that some already known hairy rodent, hitherto classified as an entirely different species, might turn out to be the lost caste of the naked mole rat?

There are, after all, precedents for this kind of thing. Locusts, for instance. Locusts are modified grasshoppers, and they normally live the solitary, cryptic, retiring life typical of a grasshopper. But under certain special conditions they change utterly—and terribly. They lose their camouflage and become vividly striped. One could almost fancy it a warning. If so, it is no idle one, for their behaviour changes too. They abandon their solitary ways and gang together, with menacing results. From the legendary biblical plagues to the present day, no animal has been so feared as a destroyer of human prosperity. They swarm in their millions, a combined harvester thrashing a path tens of miles wide, sometimes travelling at hundreds of miles per day, engulfing 2,000 tons of crops per day, and leaving a wake of starvation and ruin. And now we come to the possible analogy with mole rats. The difference between a solitary individual and its gregarious incarnation is as great as the difference between two ant castes. Moreover, just as we were postulating for the 'lost caste' of the mole rats, until 1921 the grasshopper Jekylls and their locust Hydes were classified as belonging to different species.

But alas, it doesn't seem terribly likely that mammal experts could have been so misled right up to the present day. I should say, incidentally, that ordinary, untransformed naked mole rats are sometimes seen above ground and perhaps travel farther than is generally thought. But before we abandon the 'transformed reproductive' speculation completely, the locust analogy does suggest another possibility. Perhaps naked mole rats do produce transformed reproductives, but only under certain conditions—conditions that have not arisen in recent decades. In Africa and the Middle East, locust plagues are still a menace, just as they were in biblical times. But in North America, things are different. Some grasshopper species there have the potential to turn into gregarious locusts. But, apparently because conditions haven't been right, no locust plagues have occurred in North America this century (although cicadas, a totally different kind of plague insect, still erupt regularly and, confusingly, they are called 'locusts' in colloquial American speech). Nevertheless, if a true locust plague were to occur in America today, it would not be particularly surprising: the volcano is not extinct; it is merely dormant. But if we didn't have written historical records and information from other parts of the world it *would* be a nasty surprise because the animals would be, as far as anyone knew, just ordinary, solitary, harmless grasshoppers. What if naked mole rats are like American grasshoppers, primed to produce a distinct, dispersing caste, but only under conditions which, for some reason, have not been realized this century? Nineteenth-century East Africa could have suffered swarming plagues of hairy mole rats migrating like lemmings above ground, without any records surviving to us. Or perhaps they *are* recorded in the legends and sagas of local tribes?

p. 228 … a hymenopteran female is more closely related to her sisters than she is to her offspring.

The memorable ingenuity of Hamilton's 'relatedness' hypothesis for the special case of the Hymenoptera has proved, paradoxically, an embarrassment for the reputation of his more general and fundamental theory. The haplodiploid relatedness story is just easy enough for anyone to